

ECA3102

Nanoelectronics

Exp3: Kronig Penney Model

Aim: To simulate the **Kronig–Penney model** and study the formation of **allowed energy bands and forbidden band gaps** in a periodic crystal potential.

Case Study 1: Simulate the output for Triangular well

Aim: To simulate the **electronic energy states of a triangular quantum well** and analyze the effect of the triangular potential profile on quantum confinement. The study aims to understand the distribution of discrete energy levels and their significance in nanoscale semiconductor devices.

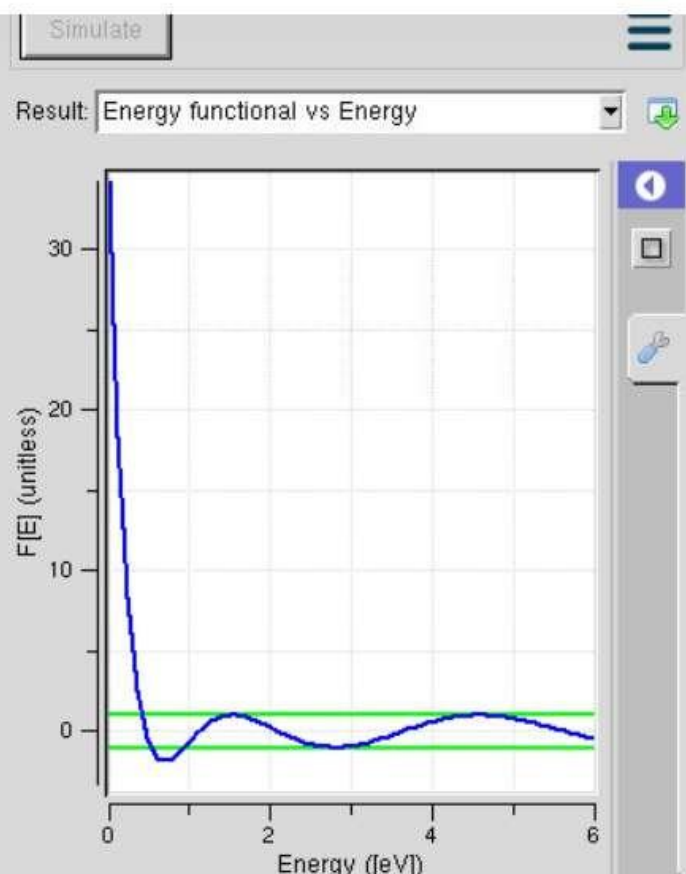
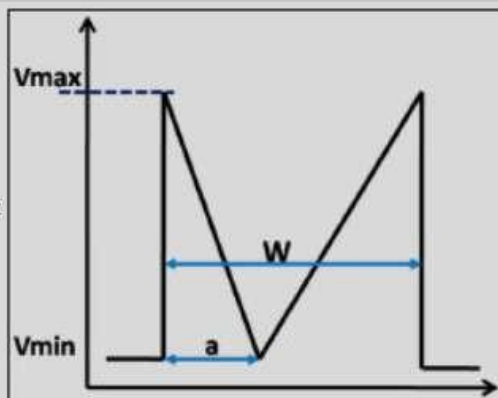


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Periodic Potential Details

Potential type | Energy Details | Well Geometry

Select the type of periodic potential: Triangular Well



2 results Clear One Clear All

Simulation = #2

All ▶ Select the type of periodic potential = Tri

Storage (manage)

0% of 10.000004GB



780 × 600

Case Study 2: Simulate the output for Sinusoidal Potential

Aim: To simulate the **electronic energy states in a sinusoidal potential** and analyze how a periodic sinusoidal potential influences the formation of energy bands and electron behavior. The study aims to understand the effect of periodic potentials on the electronic properties of semiconductor nanostructures.

